

MARSHFIELD, MA, USA

WMO: 722256

Lat: 42.098N

Lon: 70.672W

Elev: 3

StdP: 101.28

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 64774

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.5	-10.1	-22.2	0.5	-9.9	-19.9	0.6	-8.0	11.3	5.2	9.5	3.9	3.2	260	0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	32.3	23.9	30.1	22.8	28.0	21.9	25.1	29.7	24.2	28.2	23.4	26.7	3.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	18.6	27.3	22.8	17.6	26.0	22.4	17.1	25.5	76.4	29.8	72.8	28.5	69.5	26.7	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.7	7.6	6.8	-16.5	34.8	2.1	1.5	-18.1	35.9	-19.3	36.8	-20.5	37.7	-22.0	38.8
			-17.5	26.7	1.8	1.0	-18.8	27.5	-19.8	28.1	-20.9	28.7	-22.2	29.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	-0.5	0.3	3.4	8.4	14.1	18.6	22.6	21.5	17.8	12.4	6.8	2.4	(6)
(7)		DBStd	9.10	5.42	5.00	4.56	4.39	4.68	4.12	3.23	3.28	4.05	4.38	4.48	4.99	(7)
(8)		HDD10.0	1288	327	275	212	81	12	0	0	0	0	25	116	240	(8)
(9)		HDD18.3	3147	583	506	463	299	148	47	5	8	58	190	346	493	(9)
(10)		CDD10.0	1548	2	2	8	33	139	257	389	356	236	99	21	5	(10)
(11)		CDD18.3	366	0	0	1	2	18	54	136	106	43	7	0	0	(11)
(12)		CDH23.3	2711	0	0	2	24	162	403	1065	766	262	27	0	0	(12)
(13)		CDH26.7	827	0	0	0	7	52	128	355	216	66	3	0	0	(13)
(14)	Wind	WSAvg	3.0	3.3	3.3	3.6	3.4	3.0	2.8	2.6	2.4	2.6	2.9	3.1	3.1	(14)
(15)	Precipitation	PrecAvg	1290	107	95	138	113	101	93	89	103	105	117	105	120	(15)
(16)		PrecMax	1544	179	202	412	236	313	320	216	165	221	323	187	209	(16)
(17)		PrecMin	995	38	51	11	29	31	3	40	24	26	16	24	38	(17)
(18)		PrecStd	166	37	42	87	55	62	80	43	45	58	74	47	51	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.3	17.4	21.2	27.2	31.4	33.0	33.9	33.6	32.2	27.1	21.0	17.4	(19)
(20)			MCWB	14.8	13.8	13.0	17.7	20.1	25.0	24.9	25.6	24.5	20.6	17.9	15.7	(20)
(21)		2%	DB	13.0	12.9	16.3	22.1	27.6	30.0	32.3	31.0	28.0	22.9	17.9	15.1	(21)
(22)			MCWB	10.8	10.1	11.3	15.5	19.5	22.3	24.1	23.6	22.3	19.4	15.1	13.1	(22)
(23)		5%	DB	10.9	10.9	12.9	17.9	24.0	27.6	30.3	28.9	26.3	21.2	16.3	12.7	(23)
(24)			MCWB	9.0	8.5	9.4	13.1	18.4	21.0	23.1	22.6	21.2	18.3	12.9	10.8	(24)
(25)		10%	DB	7.5	7.5	10.1	15.9	21.4	25.2	28.6	27.4	24.0	18.9	13.9	10.2	(25)
(26)			MCWB	5.3	4.9	7.0	11.3	16.6	19.2	22.4	22.0	20.7	16.4	11.1	8.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.0	14.6	15.3	19.4	22.6	25.2	26.3	26.1	25.1	22.5	18.8	16.8	(27)
(28)			MCDB	16.3	16.5	17.9	22.4	28.8	31.7	32.6	32.1	29.4	24.4	20.3	17.2	(28)
(29)		2%	WB	12.2	11.3	12.5	16.8	21.0	23.5	25.2	25.0	23.7	20.5	16.6	13.8	(29)
(30)			MCDB	13.2	12.6	15.1	20.9	25.9	28.2	30.3	28.7	27.0	22.3	17.7	14.8	(30)
(31)		5%	WB	8.4	8.0	10.2	14.4	19.4	22.4	24.2	24.1	22.7	18.7	13.9	11.3	(31)
(32)			MCDB	9.6	9.8	13.4	17.7	23.1	26.0	28.4	27.5	25.3	20.9	16.0	12.5	(32)
(33)		10%	WB	5.6	5.2	7.3	12.0	17.6	21.2	23.5	23.1	21.4	16.9	12.0	8.5	(33)
(34)			MCDB	7.4	7.2	9.2	14.4	20.0	24.0	26.8	25.9	23.7	18.4	13.3	9.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.0	8.5	7.6	9.0	9.1	8.9	9.1	9.2	9.0	9.0	8.4	7.7	(35)
(36)			MCDBR	11.1	11.7	12.3	13.4	13.6	11.7	11.6	10.8	10.9	10.4	10.7	10.3	(36)
(37)			MCWBR	10.5	9.2	8.5	8.2	7.0	6.3	5.2	5.1	6.6	7.2	8.5	9.8	(37)
(38)		5% WB	MCDBR	10.7	11.3	11.5	11.9	12.5	10.7	9.8	9.4	9.3	9.6	9.4	9.9	(38)
(39)			MCWBR	10.6	10.0	9.1	8.8	7.1	6.7	5.2	5.0	6.3	7.7	8.9	10.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.300	0.308	0.329	0.371	0.407	0.458	0.479	0.437	0.377	0.340	0.315	0.301	(40)	
(41)		taud	2.515	2.503	2.489	2.395	2.337	2.231	2.178	2.288	2.437	2.540	2.571	2.568	(41)	
(42)		Ebn at Noon	854	904	920	898	869	823	800	825	857	855	832	825	(42)	
(43)		Edh at Noon	71	84	96	114	124	138	144	126	101	81	68	63	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.74	2.62	3.64	4.70	5.28	5.69	5.94	5.34	4.30	2.89	1.94	1.46	(44)	
(45)		RadStd	0.12	0.24	0.26	0.51	0.54	0.53	0.35	0.24	0.29	0.24	0.14	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	+44

Nomenclature: See separate page